



PIONEER JUNIOR COLLEGE
JC PRELIMINARY EXAMINATIONS 2011

**Candidate
Name**

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**Civics
Group**

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**INDEX
NUMBER**

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H2 BIOLOGY

9648 / 03

PAPER 3 APPLICATION PAPER

20 Sept 2011

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Read the instructions on the answer sheet very carefully.

This document consists of 2 sections:

Section A

Answer **ALL** questions.

Answers are to be written in the spaces provided.

Section B

Answer only **ONE** question.

Write your answer on the foolscap paper provided.

Do not open this booklet until you are told to do so.

For Examiner's Use	
Section A	
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Section B	
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TOTAL	/ 77

This document consists of **16** printed pages (inclusive of this page)

[Turn Over

Section A: Structured [57 marks]
Answer all questions in the spaces provided.

Question 1

Diabetes mellitus is a human disease condition in which the β cells of the islets of Langerhans in the pancreas do not produce sufficient [insulin](#). A way to treat diabetic patients is to inject purified recombinant human insulin, produced using genetically engineered bacteria, into their bloodstream.

Insulin is a relatively small protein comprising two polypeptides, one of 21 amino acids (the A chain) and one of 30 amino acids (the B chain). Trinucleotides representing all the codons are synthesized and joined together to form two artificial genes for the respective chains.

Fig. 1.1 shows the two artificial genes, including the positions of the recognition sequences of two restriction enzymes, *Pst*I and *Eco*RI, and distances (in base pairs) between the sites.

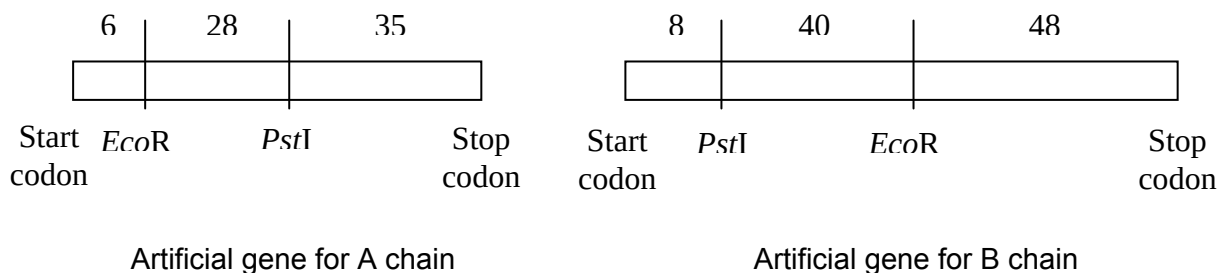


Fig. 1.1

- (a)** Apart from its small size, state another characteristic of recombinant insulin which makes it suitable to be produced using genetically engineered bacteria. [1]

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Each artificial gene is cloned into a separate bacterial plasmid pHCI18, using the restriction enzyme *HindIII*, and placed under the control of a strong *lac* promoter. **Fig. 1.2** shows the plasmid map of pHCI18. It includes the positions of the recognition sequences of three restriction enzymes, *PstI*, *EcoRI* and *HindIII*, and distances (in base pairs) between the sites. The shaded regions represent two genes, one conferring resistance to the antibiotic tetracycline (tet^R) and the other coding for the enzyme β -galactosidase (*lac Z*).

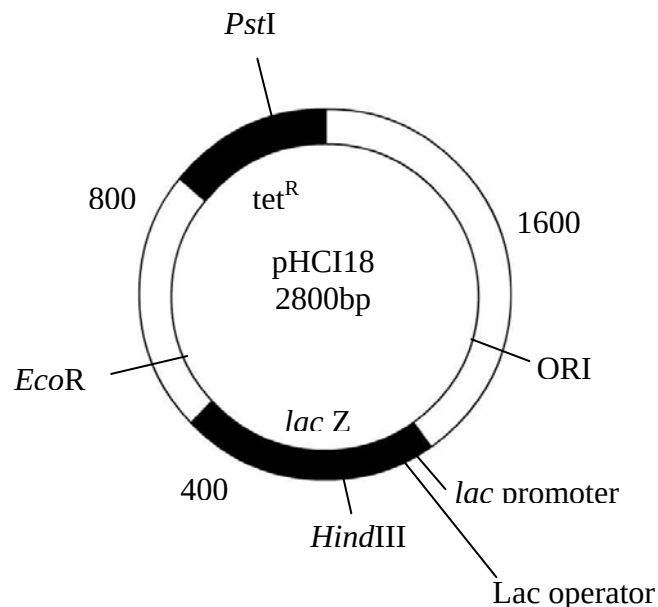


Fig. 1.2

Both recombinant plasmids containing the two artificial genes are transformed separately into *E. coli* cells. The bacterial cells from the transformation process are grown on an agar plate containing tetracycline and X-Gal.

- (b)(i) Describe how the artificial genes in the respective recombinant plasmids could be induced to undergo expression in *E. coli*. [1]

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- (ii) Explain how the selection of transformed bacterial cells could be carried out. [3]

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Once the transformed bacterial colonies have been identified, nucleic acid hybridization is carried out to further identify the colonies which contained the artificial genes.

The plasmid DNA from these colonies is then isolated and subjected to digestion by a particular restriction enzyme to determine which colonies have the artificial genes inserted in the correct orientation. A student who is unsure of which particular restriction enzyme to use carried out digestion of all the four possible recombinant plasmids numbered **A1**, **A2**, **B1** and **B2** using either enzyme *EcoRI* or *PstI*. **Fig. 1.3** shows all the results obtained.

- (c) With reference to **Fig. 1.1** and **Fig. 1.2**, complete **Fig. 1.3** by identifying the particular restriction enzyme used for digestion of the respective plasmid DNA. [2]

	Digestion of plasmid DNA with artificial gene for A chain				Digestion of plasmid DNA with artificial gene for B chain			
Plasmid	A1	A2	A1	A2	B1	B2	B1	B2
Length of fragments (in base pairs)	406	463	1234	1235	1288	1208	448	448
	2463	2406	1635	1634	1608	1688	2448	2448
Restriction enzyme used								

Fig. 1.3

- (d) With reference to **Fig. 1.1**, **Fig. 1.2** and **Fig. 1.3**,
(i) state the appropriate restriction enzyme that has to be used for the digestion of the respective plasmid DNA in order to determine which has the artificial genes inserted in the correct orientation. [1]

Restriction enzyme to digest plasmid DNA with artificial gene for A chain :

Restriction enzyme to digest plasmid DNA with artificial gene for B chain :

- (ii) explain your answers to (d)(i). [2]

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- (iii) determine which plasmids contain the artificial genes for the A and B chains inserted in the correct orientation. [1]

Plasmid with artificial gene for A chain inserted correctly :

Plasmid with artificial gene for B chain inserted correctly :

Alternatively, recombinant human insulin can be produced in eukaryotic cell lines using human insulin cDNA.

- (e) Suggest **one** advantage of using human recombinant insulin produced by eukaryotic cell lines over porcine- or bovine-derived insulin. [1]

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Apart from injection of purified recombinant human insulin, another novel way of treating diabetic patients is gene therapy, which involves the use of retroviral vectors. However, this method is still at the experimental stage.

- f)(i) Suggest two disadvantages of using retroviral vectors. [2]

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- (ii) Suggest **one** possible reason why treatment of diabetes by gene therapy is still at the experimental stage. [1]

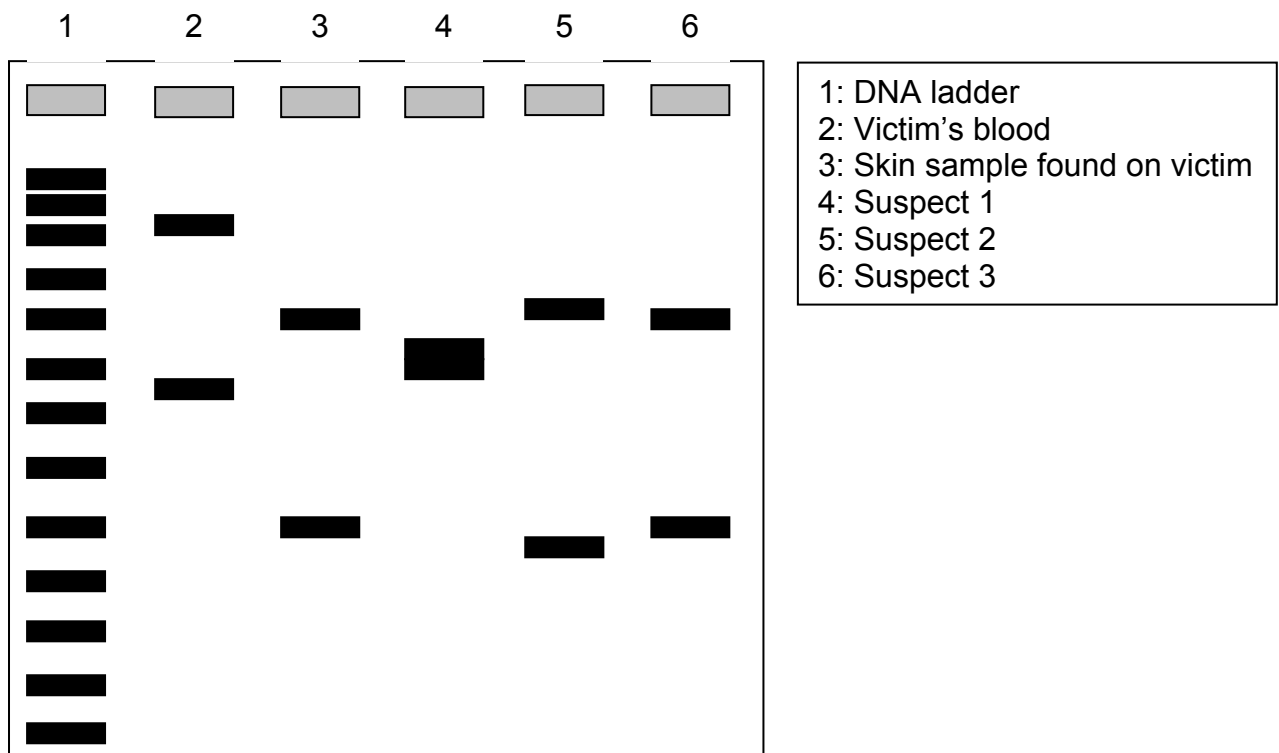
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[Total: 15 marks]

Question 2

- (a) Blood samples from 3 suspects and the victim were also taken and DNA extracted is amplified using PCR. Following PCR, the amplified DNA is cut into fragments with restriction endonucleases. One of the 13 core regions used in the forensic analysis was a short tandem repeat (STR) sequence of 4 bases **GATA**. This sequence, known as **D7S820**, is located on chromosome 7, has many alleles which differ from each other by the number of times the sequence **GATA** is repeated.

The amplified samples were loaded onto a gel and electrophoresis was performed to separate the fragments of DNA. They are separated by gel electrophoresis. The DNA is then treated with alkali and transferred onto a nitrocellulose membrane before radioactively labeled probes are added. The membrane is then washed and placed below a photographic film. The genetic fingerprint can be seen when the film is developed. A diagram of it is shown below.



- (i) Why was the DNA treated with alkali? [1]

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- (ii) What is the purpose of a DNA ladder? [1]

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(iii) Account for the different position of the bands depicted in diagram. [3]

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(iv) Why does suspect 2 have only one band while the others have 2 bands? Explain. [2]

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(v) Which individual is most likely to have committed the crime? Explain your choice. [1]

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Variable Number Tandem Repeats (VNTRs) and Short Tandem Repeats (STRs) are specific classes of human DNA polymorphisms that can be used for DNA fingerprinting.

To obtain the DNA fingerprint of an individual, three VNTR loci **1** to **3** located on chromosome 13, as shown in **Fig. 2.1**, were examined by Restriction Fragment Length Polymorphism (RFLP) analysis. The arrows mark the cleavage sites of *Eco*RI.

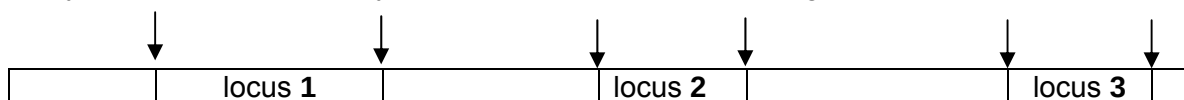


Fig. 2.1

(b) Discuss the significance of the *Eco*RI cleavage sites in this DNA fingerprinting. [2]

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Each VNTR locus contains a range of alleles defined by different numbers (**n**) of tandem repeats of a particular core sequence. The repeat sequences and their respective lengths (in bp) as well as values of **n** for the individual's VNTR loci **1** to **3** are shown in **Fig. 2.2**.

VNTR locus	Core sequence	Length (bp)	n
1	5'-TAACCGATCGGCAATG-3'	16	Between 3 and 7
2	5'-CCGATCGGCATCAGC-3'	15	Between 4 and 12
3	5'-ACCCGATCGGCA-3'	12	Between 10 and 15

Fig. 2.2

DNA fingerprinting by RFLP analysis is based on the Southern blotting technique.

- (c)(i)** Write the sequence of a 10-bp probe molecule that can be used to detect all three VNTR loci **1** to **3**. [1]

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- (ii) Outline the experimental procedure, after the action of *EcoRI*, of obtaining a DNA fingerprint. [3]

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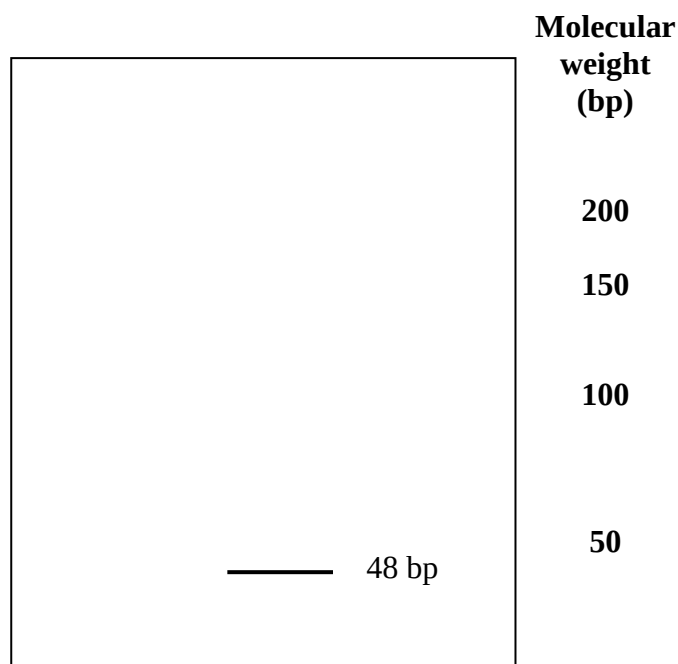
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- (d) Draw the expected results on a Southern blot below, indicating the molecular weight of each expected band. One of the expected bands is shown. [1]



[Total: 15 marks]

Question 3

Scientists discovered a protein in milk. This protein had antibacterial properties. They extracted the protein and determined its amino acid sequence. This information allowed them to make the gene for the protein. A vector was used to insert this gene into rice cells, together with a marker gene.

The scientists hope to use the genetically modified rice to make drinks that would combat diarrhoea. Diarrhoea is often caused by bacterial infections, and is a major killer of children worldwide. Many tests, however, would need to be carried out before the rice drinks could be sold to consumers.

The scientists have been given permission to plant a trial field of the genetically modified rice. They were told, however, that the trial field had to be a long way from any commercial rice farm

Use information from the passage and your own knowledge to answer the following questions

- (a) (i) A marker gene was used in genetically modifying the rice (line 4). Explain why. [2]

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- (ii) Many tests would be needed before the rice drinks could be sold to consumers (lines 7 and 8). Give **two** reasons why these tests would be needed. [2]

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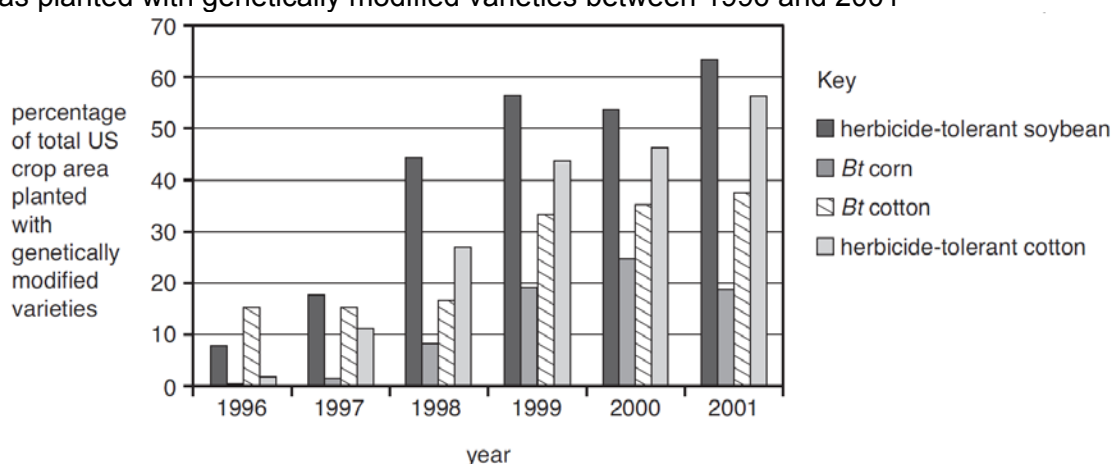
- (iii) The genetically modified rice will be grown a long way from any commercial rice farms (lines 10 and 11). Explain why this is important. [2]

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Figure 3.1 shows the percentage of the total US crop area for soybean, corn and cotton that was planted with genetically modified varieties between 1996 and 2001



- (b) (i) Describe the changes in the percentage of the crop area planted with *Bt* cotton shown in **Fig. 3.1** [2]

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- (ii) Describe how the *Bt* gene from *Bacillus thuringiensis* may be inserted into cotton plant cells. [2]

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- (c) Insecticide use and yield in India were compared for *Bt* cotton hybrid (*XBt*), the same hybrid X but without the *Bt* gene (*X₋*), and a different hybrid widely grown in that particular locality (*Y*). This process was repeated at more than 150 locations. **Table 3.1** shows the results

hybrid	<i>XBt</i>	<i>X₋</i>	<i>Y</i>
mean number of sprays against insects that eat the cotton	0.6	3.7	3.6
mean number of sprays against sap sucking insects	3.6	3.5	3.5
yield / kg ha ⁻¹	1500.0	830.0	800.0

Fig. 3.1

- (i) Explain the purpose of including hybrids *X₋* and *Y* in the study. [2]

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- (ii) State **three** conclusions that can be drawn from the data shown in **Table 3.1**. [3]

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[Total: 15 marks]

Question 4 (Planning Question)

You are required to plan, but not carry out, an investigation on the effect of temperature on the rate of photosynthesis of *Elodea*.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

1. Elodea (aquatic plant)
2. 0.1M Sodium hydrogen carbonate
3. pH buffer
4. Hot water supply
5. Ice cubes
6. Stopwatch
7. Measuring cylinders/syringe
8. Filter funnel
9. boiling tubes
10. thermometer
11. beaker
12. delivery tubes, rubber bung & rubber connections
13. lamp
14. retort stand

Your plan should have a clear and helpful structure to include:

- o an explanation of the theory to support your practical procedure
- o a description of the method used, including the scientific reasoning behind the method and any recommended safety diagrams
- o an explanation of the dependent and independent variables involved
- o how you will record your results and ensure that they are as accurate and reliable as possible
- o proposed layout of results tables and graphs with clear headings and labels
- o the correct use of technical and scientific terms

[Total: 12 marks]

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[illegible]

Section B: Essay [20 marks]

Write your answers to question 5 on the foolscap paper provided.

Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answer should be set out in sections (a), (b) etc., as indicated in the question.

Question 5

- a) Describe the treatment of SCID and cystic fibrosis, using viral and non-viral gene delivery systems [8]
- b) Genetic engineering applied to disease therapies can harness the unique properties of stem cells. Explain the basis of these unique properties of stem cells. [5]
- c) Discuss the social and ethical considerations for the use of gene therapy [3]
- d) Discuss the aims of the human genome project. [4]

- THE END -